

**The Impact of Post Type on User Engagement Intentions:
The Mediating Role of Perceived Control**

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Abstract

This study aims to explore the effect of content presentation formats (graphic and text vs. video) on user engagement on Xiaohongshu, with perceived control as a mediating variable. A quantitative research design was adopted, and data were collected and analyzed through Qualtrics. The results indicate that content presentation formats exert a significant effect on user engagement, and perceived control plays a partial and differentiated mediating role in this relationship. Specifically, graphic and text content generates higher levels of perceived control and specific types of user engagement than video content. The findings extend the existing literature on social media content and user engagement by verifying the important role of perceived control in explaining user responses to different content formats. The results also provide practical implications for platform operators and content creators to optimize content design, improve user experience, and enhance user engagement.

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1. Introduction

Xiaohongshu (XHS) has developed into a major lifestyle-sharing and social commerce platform in China. According to QianGua Data (2024), XHS has a large young user base, with post-1995 and post-2000 users accounting for a substantial proportion of active users, and its content ecosystem is highly driven by user-generated content. On the platform, image-text posts and short videos coexist as two important forms of content presentation, both of which are widely used in lifestyle sharing, product discovery, and consumption-related decision-making.

Previous studies have shown that social media engagement is not merely a behavioral outcome, but also involves users' cognitive, emotional, and behavioral responses to content (Hollebeek et al., 2014). Research on social media posts further suggests that content format and media richness, such as the use of text, photos, and videos, can influence users' clicking, liking, commenting, and sharing behaviors (Moran et al., 2020). However, while existing research has examined the relationship between content format and user engagement, limited attention has been paid to the psychological mechanisms that may explain engagement differences between image-text and short-video posts on XHS.

Exploring such psychological mechanisms is important because users' engagement intentions may not be shaped by post type alone. Users may be more willing to like, save, comment on, or share a post when the content helps them understand information, evaluate alternatives, and feel more capable of making consumption-related decisions. In this sense, perceived control is a meaningful mediator, as prior research has shown that perceived

behavioral control is closely associated with behavioral intention (Ajzen, 1991), and perceived control has also been linked to purchase intention in online shopping contexts (Li et al., 2018).

Therefore, the present study examines the association between post type, namely image-text posts versus short videos, and user engagement intentions, with perceived control as a mediating variable. Clarifying this mechanism carries both theoretical and practical value. Theoretically, this study extends social media content psychology research by shifting attention from observable engagement outcomes to the psychological processes through which content formats influence users. Practically, the findings may provide evidence-based suggestions for content creators and platform operators to refine content strategies, enhance users' sense of control, and improve the overall user experience on XHS.

2. Literature Review

2.1 User Engagement on Social Media

User engagement intentions in digital platform environments refer to users' conscious and proactive psychological tendencies to interact with platform content. Grounded in the three-tiered behavioral framework proposed by Ma, Zhao, and Gao — which categorizes engagement as consuming, participating, and producing — this construct is operationalized into five specific behavioral dimensions in the present study: liking, sharing, commenting, watching/reading, and saving based on Ma, Zhao, & Gao (2025). These five dimensions are systematically categorized into two overarching types based on their underlying motivational logic: social-oriented behaviors and utilitarian behaviors. Social-oriented behaviors comprise liking, sharing, and commenting, which reflect users' needs for interpersonal interaction, emotional expression, and social connection with other platform users. Utilitarian behaviors include watching/reading and saving, which represent users' goal-driven actions to obtain targeted information, process content

deeply, and store useful materials for future reference. This dual classification fully captures the multifaceted nature of user engagement on content-centered platforms, as it distinguishes between socially motivated interactive behaviors and practically motivated information-processing behaviors.

On hybrid content platforms such as Xiaohongshu, user engagement holds distinctive significance that extends far beyond passive entertainment consumption. Users on this platform engage with content primarily for information exchange, decision-making support, and personalized content curation, for example, users frequently save posts to retain practical guidance for subsequent life choices or consumption decisions; this perspective finds strong support in the concept of 'Utilitarian Value' consistent with Ma, Zhao, & Gao (2025). As they argue in both the Abstract and Introduction of their study, short video users are motivated to engage not just for entertainment, but to satisfy 'utilitarian needs' — specifically citing the pursuit of information and the learning of practical skills — as well as to consume algorithmically recommended 'personalized content'. Sustained and meaningful user engagement in information-seeking scenarios maintains the vitality and practical value of the platform's information ecosystem. From the perspective of information diffusion, knowledge circulates as a catalytic process where ideas propagate through the network; when these ideas reach individuals, they are recombined and synthesized to generate novel insights, which are then disseminated back into the system (Lambiotte & Panzarasa, 2009). As continuous user interaction ensures the effective circulation and application of high-quality information resources, this dynamic mechanism underpins the resilience of the digital knowledge environment. The present study focuses exclusively on user engagement within the information-seeking context, rather than e-

commerce-oriented transactional behaviors, to clarify the inherent mechanisms of content interaction in information acquisition scenarios.

Content quality has long been recognized as a key factor that captures users' attention and promotes their engagement with platform content. Drawing on the IS Success Model (DeLone, W. H. & McLean, E. R. 2003), high-quality information is a prerequisite for user satisfaction and system usage, suggesting that the quality of content fundamentally drives user interaction. However, the format through which content is delivered — specifically the image-textual mode and short video mode investigated in this study — fundamentally reshapes the ways users process, perceive, and interact with content. According to the cognitive model of media choice, rich media (such as video) high in social presence may induce distraction and reduce the ability to process complex information, whereas lean media (such as image-text) allow for greater reviewability and systematic processing (Robert & Dennis, 2005). Crucially, this suggests that differences in content delivery format alter users' cognitive experiences and behavioral choices, even when the informational value remains consistent. Consequently, these format differences directly influence users' engagement intentions across all five measured dimensions.

2.2 Media Affordances and Content Formats

This study investigates two core content formats as the independent variables in a between-subjects A/B questionnaire experiment: the image-textual mode and the short video mode. The choice to contrast these two formats is pivotal because they represent fundamentally different mechanisms of information encoding and consumption. While the image-textual mode typically requires active cognitive effort — potentially fostering the deep processing necessary for information synthesis and knowledge creation — the short video mode leverages dynamic

audio-visual stimuli to lower cognitive barriers, thereby accelerating the speed of information diffusion. By examining these formats, this research aims to determine how the medium itself influences the trade-off between the efficient circulation of information and its meaningful application within the ecosystem.

From the perspective of media affordances, this study focuses on what each format enables users to do rather than its inherent properties. As argued by Peng (2022), the affordances approach shifts the focus from static technical features to the action possibilities offered by the technology. Short videos offer high media richness with strong sensory cues, creating an immersive and vivid experience (Chen & Peng, 2023), yet their linear, flowing nature prevents users from controlling the pace of information intake. According to the cognitive model of media choice, unlike lean media (e.g., text) that allow for reviewability, rich media (e.g., video) are often characterized by real-time, linear transmission, which imposes a fixed pace on users and reduces their ability to regulate the cognitive processing flow (Robert & Dennis, 2005). In contrast, static image-textual content allows users to independently regulate their information processing speed and navigate content freely. This characteristic aligns with the 'reviewability' afforded by lean media, which enables users to control the tempo of cognitive elaboration (Robert & Dennis, 2005). This distinction further resonates with the Uses and Gratifications perspective, which suggests that users are active agents who select media to fulfill their specific psychological and social needs (Katz et al., 1973). In the context of this study, different content formats (image-textual vs. short video) may serve as distinct tools to satisfy these varying needs. Perceived control reflects individuals' subjective judgment of their ability to master and regulate information processing procedures, which further shapes their cognitive motivation and behavioral tendencies (Jewell & Kidwell, 2005). In interactive media contexts, higher perceived

control reduces perceived intrusiveness and promotes more positive user responses toward content (Li & Yin, 2021). Users select content formats according to their specific needs, and the image-textual mode provides more effective support for information-seeking goals than the passive sensory richness of short videos.

2.3 The Need for Autonomy and Perceived Control

Perceived control in digital contexts refers to users' subjective sense of agency over the medium and information flow (Liu, 2003), specifically manifested in their ability to adjust consumption pace, navigate freely, and determine the depth of their engagement with content. This construct has been empirically validated in interactive media environments, such as social media and streaming platforms, where it significantly influences user attitudes and behavioral intentions (Li & Yin, 2021). Rooted in the Theory of Planned Behavior (TPB), perceived behavioral control is a foundational construct for predicting individual behavior, capturing users' perceived capacity to exercise influence over the behavioral context (Ajzen, 1991). In digital environments, this construct is often operationalized as active control, referring to users' ability to exert agency over the information flow and medium, thereby shaping their interactive experience (Liu, 2003).

Perceived control is closely tied to the concept of autonomy, which is defined as the need to experience one's actions as volitional and self-endorsed (Ryan & Deci, 2000). As a core psychological need outlined in Self-Determination Theory (SDT), autonomy is essential for facilitating optimal functioning, behavioral self-regulation, and personal well-being. Humans possess an inherent motivation to act volitionally and feel in control of their own experiences, and this need translates directly to digital consumption contexts. In information-seeking scenarios, autonomy manifests not as abstract choice, but as concrete perceived control (Sundar

& Kalyanaraman, 2004), a concept rooted in the psychological need for autonomy (Ryan & Deci, 2000). This refers specifically to the ability to decide when, how, and at what pace to engage with content, without being constrained by the medium's structure (Liu, 2003).

While the structural differences between formats are evident, the theoretical significance of this distinction lies in its impact on cognitive resource allocation. In the context of information seeking, the primary goal is the acquisition and processing of knowledge. User-paced formats (image-text posts) are not merely convenient; they are cognitively necessary. By allowing users to pause, reread, and skip, these formats align with the natural rhythm of human cognition, enabling the conservation of cognitive resources for deep processing. Conversely, system-paced formats (short videos) force users into a state of continuous attention capture. The significance of this constraint is that it shifts the user's cognitive focus from "information analysis" to "real-time tracking," potentially overwhelming working memory. Therefore, the higher perceived control offered by graphic posts is not just a preference, but a functional requirement for effective information assimilation.

The affordances of content formats directly shape users' perceived control by either supporting or limiting their autonomy. User-paced formats, such as image-text content, grant users the flexibility to adjust their information processing rhythm and navigate content according to their own needs, thereby satisfying their need for autonomy. In contrast, system-paced formats like short videos feature a linear structure where content is pre-arranged and distributed across specific timelines by creators. Consequently, the rhythm of information delivery is dictated by the video itself rather than the user, significantly restricting their ability to exercise agency over the pace of engagement. Based on this theoretical reasoning, this study proposes:

H1: Compared with short-video posts, graphic posts tend to increase users' perceived control over information processing.

2.4 The Mediating Role of Perceived Control

According to Self-Determination Theory (SDT), autonomy-supportive environments facilitate intrinsic motivation. SDT posits that autonomy, defined as the human need to experience volition and self-endorsement in one's actions, is a fundamental driver of intrinsic motivation and psychological well-being (Ryan & Deci, 2000). Extending this principle into digital contexts, research suggests that when users perceive a high degree of control over their interactions, they are more likely to experience need satisfaction, fostering deeper engagement and persistence (Ryan, Rigby, & Przybylski, 2006).

While SDT explains the motivational underpinnings of engagement, the Theory of Planned Behavior (TPB) provides a framework for predicting the behavioral intentions that result from this motivation. Although TPB encompasses multiple determinants of behavior, including attitudes and subjective norms, this study strategically focuses on Perceived Behavioral Control (PBC) (Ajzen, 1991). Within the TPB, PBC refers to an individual's perception of the ease or difficulty of performing a behavior, determined by their assessment of available resources and potential obstacles (Ajzen, 1991). This aligns with the SDT perspective, suggesting that when users feel a high degree of perceived control, they are not only intrinsically motivated but also more likely to form strong intentions to engage in specific content interactions.

While SDT provides the motivational foundation, the specific significance of Perceived Control in this study lies in its role as the psychological bridge between media structure and utilitarian behavior. In hybrid platforms like Xiaohongshu, users seek to transform information into actionable knowledge. The theoretical deduction here is that without perceived control, the

"utilitarian value" of content cannot be realized by the user. If users feel they cannot control the flow of information (as in short videos), their engagement shifts from a goal-directed, instrumental behavior (aimed at learning) to a passive, hedonic consumption (aimed at entertainment). Therefore, Perceived Control is hypothesized to be the critical determinant that enables users to fulfill their information-seeking goals, thereby predicting their behavioral intentions. Thus, by integrating these perspectives, this study hypothesizes that perceived control positively predicts behavioral intentions across various social media actions. Specifically:

H2: Perceived control is positively associated with intentions to (H2a) like, (H2b) share, (H2c) comment, (H2d) watch/read, and (H2e) save.

Building on the preceding logic, this relationship can be connected back to the core comparison of content formats. In the context of social media consumption, the format of content (e.g., graphic vs. video) serves as a distinct set of task characteristics. According to the Theory of Planned Behavior (Ajzen, 1991), such external characteristics do not automatically translate into behavioral intentions. Instead, as posited in the theory's foundational framework, these characteristics must first be processed as 'control beliefs' — specifically, beliefs about the availability of cognitive resources and the ease of information processing. Consequently, content format is theorized to influence Perceived Behavioral Control (PBC), which reflects the user's subjective sense of agency over their engagement. This theoretical rationale supports the proposed direct effect of content format on engagement. Accordingly, the following hypothesis is advanced:

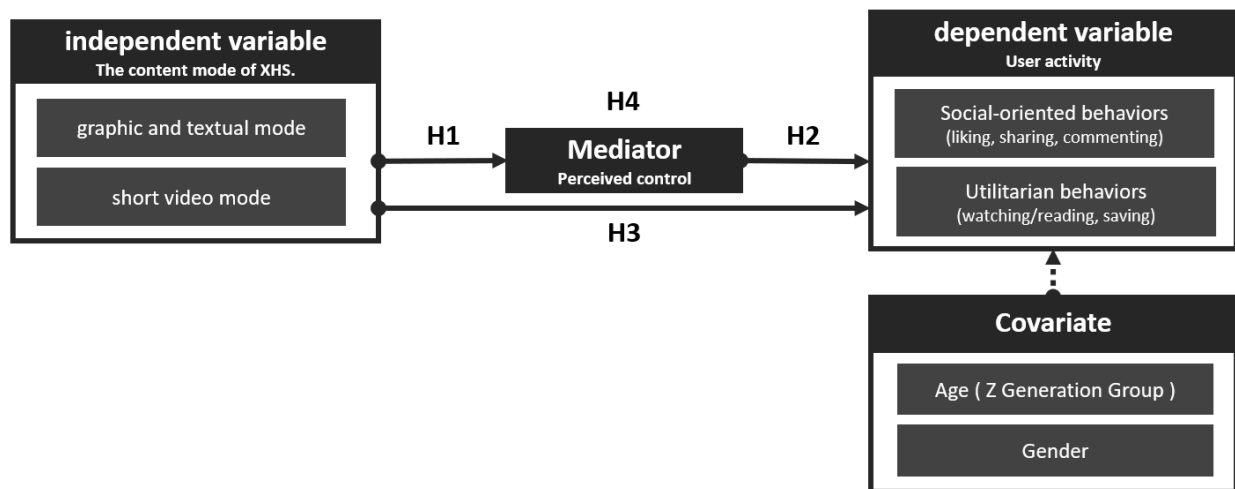
H3: Users have higher behavioural intentions to (H3a) like, (H3b) share, (H3c) comment, (H3d) watch/read, and (H3e) save graphic posts compared with short-video posts.

The prediction that graphic posts yield higher engagement intentions (H3) is significant because it highlights a fundamental tension in digital ecosystems: the trade-off between distribution efficiency (favored by videos) and information depth (favored by graphics). The theoretical contribution of this deduction is that it identifies "Perceived Control" as the mechanism that safeguards the depth of information processing in an increasingly fast-paced digital environment. By validating this mediated pathway, this study demonstrates that the medium's structure does not merely deliver content but actively shapes the user's capacity for knowledge retention and application. Because we established that graphic posts provide higher perceived control (H1), and perceived control in turn drives engagement intentions (H2), it logically follows that graphic posts will generate more favorable engagement intentions overall through this mechanism. This forms a clear mediating pathway. Accordingly, the following hypotheses are advanced:

H4: Perceived control mediates the effect of post type on the intention to (H4a) like, (H4b) share, (H4c) comment, (H4d) watch/read, and (H4e) save.

Figure 1.

Research Framework concept model



3. Method

3.1 Participants and Procedures

This study recruited adult Generation Z users of Xiaohongshu (XHS) as research participants. Generation Z users were generally defined as those born between 1995 and 2010; however, only participants who confirmed that they were at least 18 years old and had prior experience using XHS were eligible to participate in the study. A between-subjects experimental design was adopted, and participants were randomly assigned to one of two conditions: the image-text post condition or the short-video post condition.

Data were collected through an online questionnaire. A total of 224 responses were collected, including 15 pilot responses. After removing the pilot responses, 209 formal responses remained. Among these formal responses, 39 were excluded because they were invalid or did not meet the qualification criteria, leaving 170 valid responses for the final analysis. The exclusion rate among formal responses was 18.66%. Among the valid participants, 80% were female, and the mean and modal age were both 25 years. The average daily usage time of XHS was 3.5 hours, and the modal daily usage time was 4 hours. The image-text condition included 88 participants, while the short-video condition included 82 participants.

Before the formal survey, participants read the informed consent statement and confirmed that they were at least 18 years old. They were then randomly assigned to view the corresponding stimulus materials. After viewing the stimuli, participants completed scales measuring perceived control, engagement intentions, and content quality perception. Finally, demographic information, including gender, age, and daily XHS usage time, was collected.

3.2 Stimuli

The stimulus materials consisted of two types of social media posts: image-text posts and short-video posts. Specifically, the image-text condition included three image-text posts, while

the short-video condition included three short-video posts. The stimuli were selected to represent common consumption-related content formats on social media platforms.

To ensure comparability between the two experimental conditions, the image-text and short-video stimuli were matched as closely as possible in terms of topic relevance, information richness, content tone, and consumption-related context. The image-text stimuli presented information mainly through static visuals and written captions, whereas the short-video stimuli presented information through moving images, subtitles, and audio-visual elements. Participants in the image-text condition viewed image-text stimuli, while participants in the short-video condition viewed short-video stimuli. Grouped screenshots of all six stimulus materials are provided in the Appendix.

3.3 Measures

Perceived control was measured using a 7-point Likert scale, ranging from 1 = strongly disagree to 7 = strongly agree. The scale included four items assessing whether participants felt that they could browse or watch the post content at their own pace, control how they processed the content, extract the specific information they needed, and pause, replay, or skip the content according to their own preference. Higher scores indicated a stronger sense of perceived control. Cronbach's α was 0.831 in the image-text condition and 0.898 in the short-video condition.

User engagement intentions were measured using a 5-point Likert scale, ranging from 1 = very unlikely to 5 = very likely. The scale measured participants' likelihood of liking, sharing, commenting on, reading or watching, and saving the post for future reference. Higher scores indicated stronger engagement intentions. Cronbach's α was 0.873 in the image-text condition and 0.764 in the short-video condition.

Content quality perception was included as a control variable. It was measured using a 5-point Likert scale, ranging from 1 = strongly disagree to 5 = strongly agree. The two items assessed whether the post provided sufficient information and whether it contained necessary and useful information. Cronbach's α was 0.670 in the image-text condition and 0.671 in the short-video condition.

Gender, age, and daily usage time of XHS were included as covariates. All measurement items are listed in the Appendix.

3.4 Data analysis

This study's statistical analyses were performed in the Python programming language on the Google Colaboratory (Colab) platform. The packages used were Pandas, NumPy, SciPy, Pingouin and Statsmodels, which were used for cleaning and descriptive statistics, reliability tests, correlation and hypothesis testing. The analysis procedure adhered to quantitative research standards. The five aspects of user engagement were considered as different dependent variables and were investigated separately, instead of creating a composite score. This allows us to specifically capture the psychological effects of each behaviour, rather than blending them together.

3.4.1 Data Preprocessing

In data preprocessing, we excluded invalid responses based on response time, logical consistency, and response rate for quality control. Valid responses were coded into two groups: Group A (image and text posts, $n = 82$) and Group B (short-video posts, $n = 88$). Scale items were automatically identified for each condition using regular expressions. Listwise deletion was applied to deal with missing data and maintain the integrity of the analyses.

3.4.2 Reliability Analysis

Reliability analysis was conducted on the internal consistency of each scale using Cronbach's α coefficient. The standards were that a value of $\alpha > 0.70$ was considered acceptable and $\alpha > 0.80$ was considered good. The overall reliability was calculated for the total sample rather than grouped. The findings showed acceptable reliability for perceived control and engagement intention scales, with the scale for content quality perception close to acceptable reliability, suggesting that the measurement instrument was reliable.

Willingness to participate is defined as the self-reported intention of the user to engage in interactive activities with social media content, such as like, share, comment, read or view, and save. It is the extent to which an individual is willing or likely to engage in a specific interactive behavior, although it is not the actual interactive behavior. In the present study, these five engagement intentions were defined as willingness to participate since they reflect the proactive psychological state of the user to engage with social media content, which aligns with the definition of engagement intention in the social media and communication literature.

Table 1

Cronbach's α Reliability Test Results for Scales (Total Sample)

Scale	Cronbach's α (Total Sample)
Perceptual Control (PC) α	0.831
Willingness to participate (Intent) α	0.873
Content Quality Perception (PCQ) α	0.670

3.4.3 Descriptive Statistics

Descriptive Statistics primarily describe the means (M) and standard deviations (SD) of the core variables (perceived control, intention to participate and perceived content quality) for the two groups. The findings reveal that the image-text group (Group A) has higher scores than the

short-video group (Group B) in all the core variables, with perceived control showing the largest difference between the two groups. The standard deviations of the two groups are relatively low, which means the two groups are consistent.

Table 2

Mean and Standard Deviation of Core Variables ($M \pm SD$)

Core variable	Group A (Image & Text) $M \pm SD$	Group B (Short Video) $M \pm SD$
PC	5.82 $\pm$ 0.56	4.37 $\pm$ 0.95
Intent	3.48 $\pm$ 0.64	2.96 $\pm$ 0.48
PCQ	3.96 $\pm$ 0.46	3.40 $\pm$ 0.49

3.4.4 Preliminary Correlation Analysis and Hypothesis Testing Strategy

To test the correlation between variables, preliminary Pearson correlation analyses were performed. First, we examined the intercorrelations between the five engagement behaviors (liking, sharing, commenting, reading/viewing, and saving). These five indicators were found to be moderately positively correlated ($r = 0.396$ - 0.649), suggesting that these engagement activities are conceptually related but measurable and non-redundant. This finding justifies using each of these engagement behaviors as an independent dependent variable in the forthcoming analyses.

It should be noted that Pearson correlation was adopted to explore the relationships among variables for preliminary analyses, but not for testing hypotheses. Given the study's between-subjects experimental design with two levels of independent variable (image-text post and short video post groups), independent samples t-tests were adopted as the statistical technique for testing whether mean differences are significant. Pearson correlation cannot be used to infer causality or conduct experiments; thus, t-tests were applied to test whether there is a significant difference in perceived control and intentions of each type of user engagement when comparing different post types.

3.4.5 Hypothesis Testing

In the hypothesis testing, the study used the independent samples t-test to compare the differences in the mean under the two experimental conditions. To align with the theoretical basis and analytical needs, individual participating behavior is tested, rather than the total score. This approach successfully identifies the different effects of post type and perceived control on various participation intentions.

This study used the Bootstrap method with 5000 repeated samplings to test the mediating effect, to ensure the mediating effect of perceived control. In order to ensure the consistency of the analytical framework, this study built five separate mediating models for the five dimensions of participating behaviors, and tested for the significance of the indirect effect using the 95% bias-corrected confidence interval.

In each analysis, post type was the independent variable, perceived control was the mediating variable and each participating behavior was the independent dependent variables. This approach guarantees the theoretical validity of the study, clear interpretation of the results, and is in line with the aim of this study to explore the different psychological foundations of user participation behaviors.

4. Results: Hypothesis Testing

4.1 Hypothesis 1: Content Format and Perceived Control

Hypothesis 1 proposed that image-text posts would increase users' perceived control over information processing compared to short-video posts. A Welch's independent samples t-test (equal variances not assumed) was conducted to compare perceived control scores between the two conditions. The results revealed that participants in the image-text condition reported

significantly higher perceived control ($M = 5.82$, $SD = 0.56$) than those in the short video condition ($M = 4.37$, $SD = 0.95$), $t(168) = 12.29$, $p < .001$.

The positive and large t -statistic indicated that the mean perceived control score was substantially higher in the image-text group, while the p -value close to zero confirmed a highly significant difference between the two conditions. These findings fully support Hypothesis 1. Critically, perceived control is interpreted not as a general sense of ease in reading, but as a user's subjective agency over the pace, navigation, and manner of information processing (e.g., the ability to pause, skip, or revisit content). Thus, the results demonstrate that image-text formats grant users a stronger sense of control over their information-seeking experience compared to short videos, aligning with the hypothesized direction of the relationship.

4.2 Hypothesis 2: Perceived Control and Engagement Intentions

Hypothesis 2 proposed that perceived control (PC) is positively associated with intentions to (H2a) like, (H2b) share, (H2c) comment, (H2d) watch/read, and (H3e) save. Pearson correlation analyses were conducted to test the relationships between perceived control and each engagement dimension, with results presented in Table 3.

Table 3

Pearson Correlations Between Perceived Control and Engagement Intentions

Behavior	Correlation Coefficient (r)	p -value
Liking	.307	< .001
Sharing	.410	< .001
Commenting	.287	< .001
Watching/Reading	.401	< .001
Saving	.549	< .001

Consistent with the motivational drivers of user interaction, we categorized the five engagement behaviors into two distinct groups: social-oriented behaviors (liking, sharing, commenting) and utilitarian behaviors (watching/reading, saving). All correlation coefficients were positive and statistically significant at the $p < .001$ level. For social-oriented behaviors, perceived control showed moderate positive correlations with liking ($r = .307$), sharing ($r = .410$), and commenting ($r = .287$). For utilitarian behaviors, the associations were stronger, with perceived control correlating positively with watching/reading ($r = .401$) and most strongly with saving ($r = .549$). These results indicate that perceived control significantly predicts both social connection behaviors and goal-driven information-processing actions, providing full support for Hypothesis 2.

4.3 Hypothesis 3: Content Format and Engagement Intentions

Hypothesis 3 proposed that users would report higher behavioral intentions to like, share, comment, watch/read, and save graphic posts compared with short-video posts. To test this, independent samples Welch's t-tests (equal variances not assumed) were conducted separately for each of the five engagement dimensions.

Independent-samples t-tests revealed significant differences across all five engagement intentions (all $p < .001$).

For social-oriented behaviors, participants in the image-text condition ($n = 82$) reported significantly higher intentions than those in the short video condition ($n = 88$) for:

- **liking** ($M_{\text{graphic}} = 3.60, SD = 0.78$ vs. $M_{\text{video}} = 3.16, SD = 0.63$; $t = 3.98, p < .001$),
- **sharing** ($M_{\text{graphic}} = 3.40, SD = 0.85$ vs. $M_{\text{video}} = 2.86, SD = 0.70$; $t = 4.44, p < .001$),
- **commenting** ($M_{\text{graphic}} = 3.13, SD = 0.96$ vs. $M_{\text{video}} = 2.58, SD = 0.72$; $t = 4.21, p < .001$).

For utilitarian behaviors, image-text posts also elicited significantly higher intentions for:

- **watching/reading** ($M_{\text{graphic}} = 3.88, SD = 0.55$ vs. $M_{\text{video}} = 3.54, SD = 0.58$; $t = 3.92, p < .001$),

- *saving* ($M_{\text{graphic}} = 3.38$, $SD = 0.73$ vs. $M_{\text{video}} = 2.65$, $SD = 0.78$; $t = 6.31$, $p < .001$).

The strongest difference was observed for saving intentions (mean difference = 0.73), indicating that the format's effect was most pronounced for goal-driven, resource-curating behaviors.

Collectively, these results provide full support for Hypothesis 3, showing that image-text formats are associated with higher engagement intentions across both social and utilitarian dimensions.

Table 4

Independent Samples t-Test Results for Engagement Intentions by Content Format

Behavior	<i>Graphic Posts (A)</i>	<i>Short-Video Posts (B)</i>	<i>t</i> -value	<i>p</i> -value
	<i>M (SD)</i>	<i>M (SD)</i>		
Liking	3.60 (0.78)	3.16 (0.63)	3.98	< .001
Sharing	3.40 (0.85)	2.86 (0.70)	4.44	< .001
Commenting	3.13 (0.96)	2.58 (0.72)	4.21	< .001
Watching/Reading	3.88 (0.55)	3.54 (0.58)	3.92	< .001
Saving	3.38 (0.73)	2.65 (0.78)	6.31	< .001

4.4 Hypothesis 4: Mediating Role of Perceived Control

Hypothesis 4 proposed that perceived control (PC) mediates the effect of post type (image-text vs. short video) on each of the five engagement intentions. Separate mediation analyses were conducted using bootstrapping (5,000 resamples; pingouin mediation analysis).

Results are reported with 95% bias-corrected bootstrap confidence intervals (CI) and p-values for the indirect effects.

The results are presented below for each path coefficient:

1. Total effect (c): Post type had significant positive total effects on all five engagement behaviors: liking ($c = 0.435$, $p < .001$), sharing ($c = 0.535$, $p < .001$), commenting ($c = 0.548$, $p < .001$), watching/reading ($c = 0.342$, $p < .001$), and saving ($c = 0.728$, $p < .001$). This indicates that image-text posts consistently elicited higher overall engagement intentions than short-video posts.

2. Path a (post type $\rightarrow$ perceived control): Across all models, post type had a consistent, significant positive effect on perceived control ($a = 1.454$, $p < .001$, 95% CI [1.217, 1.692]), indicating that image-text posts enhanced users' sense of control over information processing.

3. Path b (perceived control $\rightarrow$ engagement, controlling for post type): Perceived control significantly predicted all engagement intentions: liking ($b = 0.135$), sharing ($b = 0.270$), commenting ($b = 0.115$), watching/reading ($b = 0.211$), and saving ($b = 0.367$) – all $p < .001$. Thus, higher perceived control was associated with greater willingness to engage, regardless of content format.

4. Indirect effect (ab) : The mediated effect varied across behaviors: .

- **Sharing:** indirect effect = 0.392, 95% CI [0.130, 0.702], $p = .006$ – **significant**
- **Watching/reading:** indirect effect = 0.307, 95% CI [0.134, 0.531], $p < .001$ – **significant**
- **Saving:** indirect effect = 0.534, 95% CI [0.282, 0.829], $p < .001$ – **significant**
- **Liking:** indirect effect = 0.197, 95% CI [-0.054, 0.470], $p = .140$ – **not significant**
- **Commenting:** indirect effect = 0.167, 95% CI [-0.129, 0.487], $p = .286$ – **not significant**

5. Direct effect (c'): The remaining effect of post type on each engagement behavior after controlling for perceived control. The direct effects were substantially reduced compared to the

total effects: liking ($c' = 0.238$), sharing ($c' = 0.142$), commenting ($c' = 0.382$), watching/reading ($c' = 0.035$), and saving ($c' = 0.195$). For watching/reading, the direct effect became very small, suggesting full or near-full mediation for this specific behavior.

These results provide partial support for Hypothesis 4. Post type influences engagement intentions through perceived control, but this mediating effect is behavior-specific. Specifically, perceived control fully mediated the effect of post type on watching/reading intentions, and significantly mediated the effects on sharing and saving intentions. This suggests that graphic-text posts enhance users' sense of control over information processing, thereby facilitating engagement behaviors that require deeper cognitive processing and information preservation (i.e., watching/reading, sharing, and saving). However, the mediating role of perceived control was not significant for liking and commenting intentions, implying that these behaviors are likely driven by other psychological mechanisms — such as emotional resonance or social norms — rather than by perceived control alone. Overall, while graphic-text posts consistently elicited higher engagement intentions across all five dimensions, the pathway through perceived control only explains a subset of these effects, underscoring the need to consider multiple underlying mechanisms for different types of engagement behaviors.

Table 5

Mediation Analysis Results for Perceived Control on Individual Engagement Intentions

Behavior	Total Effect (c)	Path a	Path b	Indirect Effect (ab)	Direct Effect (c')	95% CI	Indirect p-value
Liking	0.435	1.454	0.135	0.197	0.238	[-0.0543, 0.4698]	0.140
Sharing	0.535	1.454	0.270	0.392	0.142	[0.1296, 0.7016]	0.006
Commenting	0.548	1.454	0.115	0.167	0.382	[-0.1290, 0.4867]	0.286
Watching/Reading	0.342	1.454	0.211	0.307	0.035	[0.1340, 0.5312]	0.000
Saving	0.728	1.454	0.367	0.534	0.195	[0.2817, 0.8286]	0.000

Note. Coefficients are rounded to three decimal places.

5. Discussion

The primary objective of this study was to investigate how post type (image-text vs. short video) influences user engagement intentions on Xiaohongshu, with a particular focus on the mediating role of perceived control. Drawing on Self-Determination Theory (SDT) and media affordances, we proposed that image-text posts, by affording greater navigational control, would enhance perceived control, which in turn drives engagement intentions. The results provide robust support for H1, H2, and H3, and partial support for H4, offering both theoretical and practical insights.

5.1 Summary of Key Findings

Consistent with H1, participants exposed to image-text posts reported significantly higher levels of perceived control over information processing ($M = 5.82$, $SD = 0.56$) compared to those exposed to short-video posts ($M = 4.37$, $SD = 0.95$), Welch's $t(168) = 12.29$, $p < .001$, confirming that user-paced formats satisfy the psychological need for autonomy, whereas system-paced video formats restrict such control.

Supporting H2, perceived control was positively associated with all five engagement intentions, with the strongest link observed for saving behavior. This pattern suggests that perceived control is especially important when users engage in utilitarian, future-oriented information curation, rather than momentary social interactions.

In line with H3, image-text posts consistently generated higher engagement intentions across all five dimensions compared to short-video posts. This finding suggests that in information-seeking contexts, content formats that afford greater user control over pacing and navigation — even if they are less sensorially rich — can be more effective in driving engagement than more immersive but system-paced formats.

Regarding H4, the mediation analysis revealed a differentiated pattern. For sharing, watching/reading, and saving, perceived control significantly mediated the effect of post type, with the direct effects becoming non-significant, suggesting that perceived control plays a dominant mediating role. However, for liking and commenting, the mediation effect was non-significant, as indicated by bootstrap confidence intervals containing zero and p values above .05 (liking: 95% CI [-0.054, 0.470], $p = .140$; commenting: 95% CI [-0.129, 0.487], $p = .286$). Thus, the mediating role of perceived control is behavior specific rather than universal across all five engagement intentions. The indirect effect was highly significant, indicating that the advantage of image-text posts over short videos in driving user engagement is largely explained by the enhanced sense of perceived control they afford.

5.2 Theoretical Implications

This study makes several important theoretical contributions. First, it bridges a critical gap between platform design features and deep psychological needs by identifying perceived control as a key mediating mechanism for specific engagement behaviors. Our findings demonstrate that format is not a passive container but an active determinant of user experience. Specifically, image-text posts function as a high-control provision type, whereas short videos represent a low-control provision type.

Second, we found that although video is richer in sensory cues (e.g., motion, sound), the structural affordances of image-text formats — such as skimming, pausing, and non-linear navigation — prove more critical when users need to process dense or utilitarian information. This suggests that media richness should not be equated with communication effectiveness; rather, the match between affordances and user goals determines outcomes.

Third, by operationalizing five distinct engagement behaviors, this study refines the conceptualization of user engagement on social media. Engagement is not a monolithic construct. Our results show that perceived control differentially impacts engagement types: the strongest effect was on saving (utilitarian, long-term), followed by watching/reading and liking, with relatively weaker effects on sharing and commenting. This nuance suggests that user-paced formats are particularly adept at driving goal-oriented, deliberative engagement, whereas system-paced formats may be better suited for affective, immediate reactions.

Finally, this study extends Self-Determination Theory to new media contexts by demonstrating that perceived control serves as the digital manifestation of the need for autonomy. Our research provides empirical evidence that satisfying the need for autonomy via user-paced formats directly fuels intrinsic motivation and subsequent engagement behaviors.

5.3 Practical Implications

For content creators on platforms like Xiaohongshu, our findings suggest a strategic choice based on engagement goals: when the goal involves high-density, decision-oriented information (e.g., product reviews, tutorials, outfit matching). For platform designers, enhancing short videos with control-affording features — such as pause, progress bars, speed control, and non-linear navigation — could mitigate the perceived control deficit and improve engagement. For brand marketers, when engagement and retention are key objectives (e.g., saving posts for future purchase consideration), image-text posts may offer a more effective strategy than short videos.

5.4 Limitations and Future Directions

Despite the robust findings, this study has several limitations. First, the binary comparison (image-text vs. short video) does not account for hybrid or multimodal content forms

(e.g., video with text overlay, livestream clips), which are increasingly common on social media platforms. Future research should sample such hybrid formats to test the boundary conditions of our findings.

Second, the cross-sectional, self-reported design introduces potential common method bias. While random assignment to conditions supports causal claims regarding post type, the temporal stability and generalizability of the observed mediation model would benefit from replication using longitudinal or repeated-measures designs. Future research could also complement self-reported intentions with behavioral logs.

Third, the sample was predominantly young, female (80%), and from first-tier cities, limiting generalizability to older, male, or rural user populations. Future studies should stratify sampling by age, city tier, usage intensity, and behavioral patterns.

Fourth, the measure of content quality perception showed borderline reliability ($\alpha=0.67$), suggesting that future research should refine this scale or control for content quality more rigorously.

Finally, we did not examine potential moderators such as content topic (hedonic vs. utilitarian), user media literacy, or prior platform experience. Future research could incorporate these variables to further unpack when and for whom perceived control matters most. Additionally, applying computational methods (e.g., NLP for information density, computer vision for visual complexity) could provide more objective characterizations of post types.

5.5 Conclusion

In conclusion, this study demonstrates that image-text posts outperform short videos in driving user engagement intentions on Xiaohongshu, and that this effect is partially mediated by perceived control, particularly for utilitarian and information-seeking behaviors. By shifting the

theoretical lens from what media are (rich vs. lean) to what they allow users to do (control vs. passive consumption), we provide a more psychologically grounded account of user engagement in hybrid social media environments. As algorithm-driven, system-paced content continues to proliferate, satisfying users' fundamental need for autonomy through controllable formats may become an increasingly critical strategy for platforms and creators alike.

References

- Ajzen, I. (1991). The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2), 179–211. doi:10.1016/0749-5978(91)90020-T
- Hollebeek, L. D., Glynn, M. S., & Brodie, R. J. (2014). Consumer brand engagement in social media: Conceptualization, scale development and validation. *Journal of Interactive Marketing*, 28(2), 149–165. doi:10.1016/j.intmar.2013.12.002
- Li, Y., Xu, Z., & Xu, F. (2018). Perceived control and purchase intention in online shopping: The mediating role of self-efficacy. *Social Behavior and Personality: An International Journal*, 46(1), 99–106. doi:10.2224/sbp.6377
- Moran, G., Muzellec, L., & Johnson, D. (2020). Message content features and social media engagement: Evidence from the media industry. *Journal of Product & Brand Management*, 29(5), 533–545. doi:10.1108/JPBM-09-2018-2014
- QianGua Data. (2024, April 10). 2024 “Active Users” Research Report: Xiaohongshu Platform. QianGua Data.
- Ma, Z., Zhao, Y., & Gao, Q. (2025). How do social media affordances influence perceived value and short video users’ engagement behavior: A comparison between young and senior group. *International Journal of Human–Computer Interaction*, 41(24), 15353-15371.
- Lambiotte, R., & Panzarasa, P. (2009). Communities, knowledge creation, and information diffusion. *Journal of informetrics*, 3(3), 180-190.
- DeLone, W. H., & McLean, E. R. (2003). The DeLone and McLean model of information systems success: a ten-year update. *Journal of management information systems*, 19(4), 9-30.

- Robert, L. P., & Dennis, A. R. (2005). Paradox of richness: A cognitive model of media choice. *IEEE transactions on professional communication*, 48(1), 10-21.
- Jewell, R. D., & Kidwell, B. (2005). The moderating effect of perceived control on motivation to engage in deliberative processing. *Psychology & Marketing*, 22(9), 751–769.
<https://doi.org/10.1002/mar.20083>
- Chen, M., & Peng, A. Y. (2023). Why do people choose different social media platforms? Linking use motives with social media affordances and personalities. *Social Science Computer Review*, 41(2), 330-352.
- Liu, Y. (2003). Developing a scale to measure the interactivity of websites. *Journal of advertising research*, 43(2), 207-216.
- Ryan, R. M., & Deci, E. L. (2000). Self-determination theory and the facilitation of intrinsic motivation, social development, and well-being. *American psychologist*, 55(1), 68.
- Ryan, R. M., Rigby, C. S., & Przybylski, A. (2006). The motivational pull of video games: A self-determination theory approach. *Motivation and emotion*, 30(4), 344-360.
- Li, B., & Yin, S. (2021). How perceived control affects advertising avoidance intention in a skippable advertising context: A moderated mediation model. *Chinese Journal of Communication*, 14(2), 157–175. <https://doi.org/10.1080/17544750.2020.1776743>
- Katz, E., Blumler, J. G., & Gurevitch, M. (1973). Uses and gratifications research. *The public opinion quarterly*, 37(4), 509-523.
- Bigne, E., Simonetti, A., Ruiz, C., & Kakaria, S. (2021). How online advertising competes with user-generated content in TripAdvisor. *Journal of Business Research*, 123, 279–288.

- Gkikas, D. C., Tzafilkou, K., Theodoridis, P. K., & Garmpis, A. (2022). How do text characteristics impact user engagement in social media posts. *International Journal of Information Management Data Insights*, 2(1), 100067.
- Kaur, W., Balakrishnan, V., Rana, O., & Sinniah, A. (2019). Liking, sharing, commenting and reacting on Facebook. *Telematics and Informatics*, 39, 25–36.
- Li, K., Zhou, C., & Yu, X. (2021). Exploring the differences of users' interaction behaviors on microblog. *Telematics and Informatics*, 59, 101553.
- Qin, M., Qiu, S., Zhao, Y., Zhu, W., & Li, S. (2024). Graphic or short video? The influence mechanism of UGC types on consumers' purchase intention. *Electronic Commerce Research and Applications*, 65, 101402.
- Sundar, S. S. (2008). The MAIN model: A heuristic approach to understanding technology effects on credibility. *Media Psychology*, 11(4), 730–755.

Appendix

Stimulus Materials and Measurement Items

A. Stimulus Materials

The complete stimulus materials used in the experiment are provided below.

Figure A1. Screenshot of the image-text post stimulus

(a) Image-text post stimulus 1.

(b) Image-text post stimulus 2.

(c) Image-text post stimulus 3.

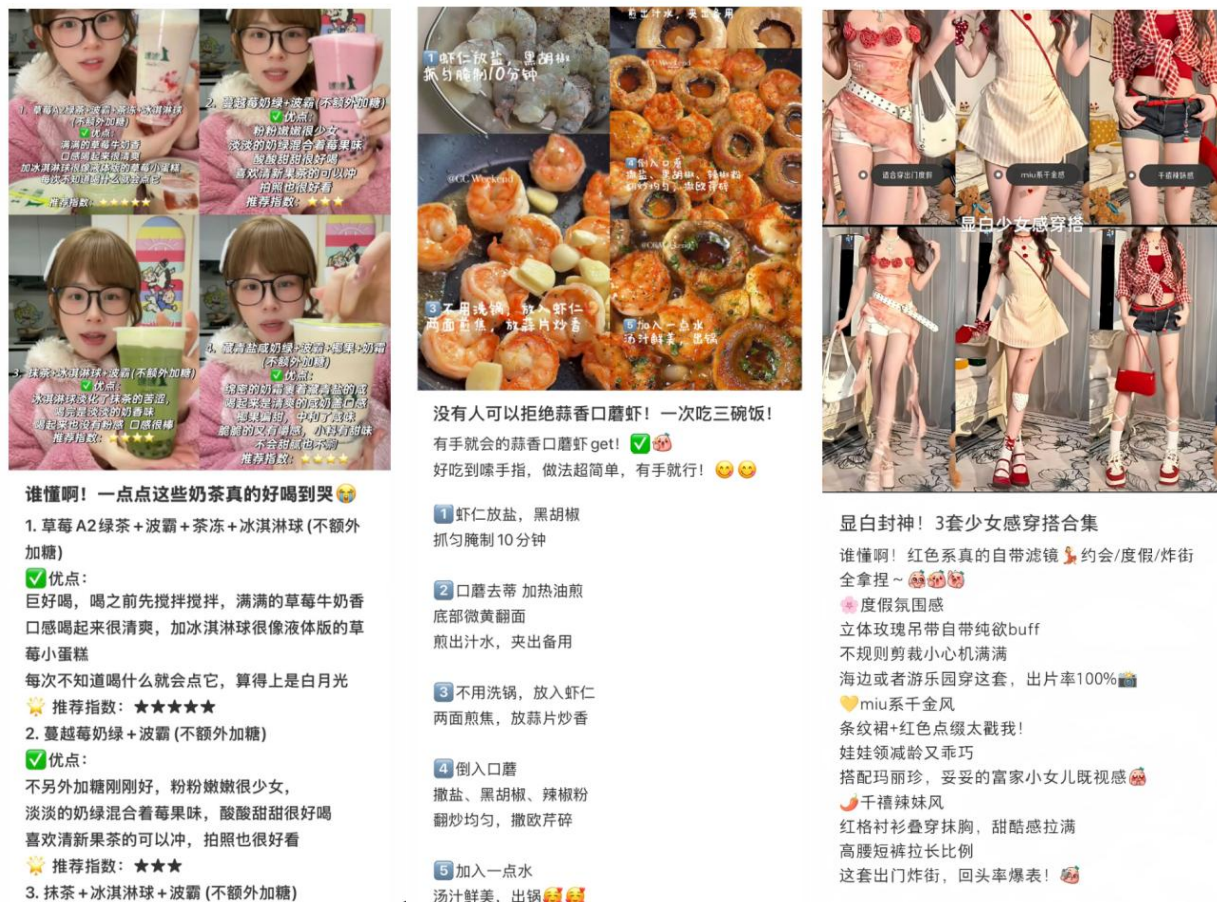


Figure A2. Screenshot of the short-video post stimulus

(a) Short-video post stimulus 1.

(b) Short-video post stimulus 2.

(c) Short-video post stimulus 3.



来源: <https://v.douyin.com/HYySIhc0elo/>



来源: <https://v.douyin.com/ZNGuW15EAuM/>



来源: <http://xhslink.com/o/1QJ019E51pT>

B. Measurement Items

Perceived control

Perceived control was measured using four items on a 7-point Likert scale, ranging from 1 = strongly disagree to 7 = strongly agree. The items were:

1. I feel that I can browse/watch the information in this post at my own pace.
2. I feel that I can control how I browse/watch the content of this post.
3. I think I can easily extract the specific information I need from this post.
4. I can conveniently pause, replay, or skip the content of this post according to my own preference.

For the image-text condition, the wording “browse” was used. For the short-video condition, the wording “watch” was used when appropriate.

User engagement intentions

User engagement intentions were measured using five items on a 5-point Likert scale, ranging from 1 = very unlikely to 5 = very likely. The items were:

1. How likely are you to like this post when you see it on social media?
2. How likely are you to share this post when you see it on social media?
3. How likely are you to comment on this post when you see it on social media?
4. How likely are you to read/watch this post when you see it on social media?
5. How likely are you to save this post for future reference when you see it on social media?

For the image-text condition, the wording “read” or “browse” was used. For the short-video condition, the wording “watch” was used.

Content quality perception

Content quality perception was measured using two items on a 5-point Likert scale, ranging from 1 = strongly disagree to 5 = strongly agree. The items were:

1. This post provides sufficient information.
2. This post contains necessary and useful information.

Covariates

The covariates included gender, age, and average daily usage time of XHS in the past week.

Gender was measured with two options: female and male. Age was measured through an open-ended response. Daily XHS usage time was measured on a scale from 0 to 8 hours.